Lecture 10

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From Portfolio Choice to Asset Pricing

- Modern finance begins with portfolio choice:
  - investors allocate wealth across risky assets
  - diversification reduces idiosyncratic risk
  - optimal portfolios lie on the efficient frontier
- But portfolio theory takes expected returns as inputs:
  - $E(R_i)$ must be assumed or estimated externally
  - the theory does not explain why returns differ across assets
- Asset pricing theory asks the equilibrium question:
  - What expected returns are required so that investors are willing to hold the existing supply of risky assets?

Portfolio Theory vs. CAPM

- Markowitz portfolio theory solves an investor's allocation problem:
  - Inputs:
    - expected returns
    - variances and covariances
  - Output:
    - efficient portfolios
    - optimal risky portfolio (with R_f)
  - Question answered: how should investors combine risky assets?
- CAPM extends portfolio theory to market equilibrium:
  - Investors hold the same optimal risky portfolio
  - All assets must be held in aggregate
  - Expected returns adjust until no investor wants to rebalance
  - Question answered: what returns must assets offer in equilibrium?

CAPM Assumptions

- CAPM is derived under a strong but transparent set of assumptions:
  - Investors are mean–variance optimizers
  - Single-period investment horizon
  - Homogeneous expectations about expected returns, variances and covariances
  - Perfect capital markets:
    - no taxes or transaction costs
    - unlimited short selling
    - borrowing/lending at the same risk-free rate
    - All assets are publicly traded and infinitely divisible
- These assumptions ensure that portfolio choice leads to a simple equilibrium pricing relation.

The Market Portfolio

- In equilibrium, the aggregate risky portfolio held by all investors must equal the supply of risky assets.
  - Because the market portfolio is:
    - Aggregation of all investors' risky portfolio
    - And investors' risky portfolio must be identical
    - Then, market portfolio will have those same weights
  - This portfolio is the market portfolio, containing:
    - All traded risky securities
    - The proportions for each stock (or other asset) must be its market value
    - Thus, investors simply have a share in the market portfolio
  - The market portfolio is:
    - fully diversified
    - the tangency portfolio from mean–variance analysis
    - the benchmark for systematic risk

From Portfolio Risk to Expected Returns

- Let the market portfolio be the value-weighted portfolio of all risky assets:

$$R_M = \sum_{i=1}^{n} w_i R_i$$

- Consider an individual security, Stock X, with return R_X and weight w_X.
- Using covariance additivity and linearity:

$$\sum_{i=1}^{n} w_i \operatorname{Cov}(R_i, R_X) = \operatorname{Cov}\Big(\sum_{i=1}^{n} w_i R_i,\ R_X\Big) = \operatorname{Cov}(R_M, R_X)$$

- Therefore, Stock X's contribution to the variance of the market portfolio is proportional to:

$$w_X \operatorname{Cov}(R_M, R_X)$$

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Reward-to-Risk Ratio and the Market Price of Risk

- Stock X's contribution to the market risk premium is: $w_X\, E(r_X)$; where $r_X = R_X - R_f$
- Stock X's contribution to the market variance is: $w_X\, \mathrm{Cov}(r_X, r_M)$
- The reward-to-risk ratio for Stock X as part of the efficient market portfolio is:

$$\frac{\text{Contribution to risk premium}}{\text{Contribution to variance}} = \frac{w_X E(r_X)}{w_X \mathrm{Cov}(r_X, r_M)} = \frac{E(r_X)}{\mathrm{Cov}(r_X, r_M)}$$

- For the market portfolio itself (the tangency portfolio):

$$\frac{\text{Market risk premium}}{\text{Market variance}} = \frac{E(r_M)}{\sigma_M^2}$$

- This ratio is called the market price of risk:
  - extra return demanded per unit of systematic variance

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Equilibrium Condition: Equal Reward-to-Risk Ratios

- Equilibrium principle:
  - if Stock X offered a better reward-to-risk tradeoff than other assets, investors would tilt portfolios toward X
  - prices would adjust until no such advantage remains
- Therefore, in the market portfolio:

$$\frac{E(r_i)}{\mathrm{Cov}(r_i, r_M)} = \frac{E(r_M)}{\sigma_M^2}$$

- Rearranging gives the fair risk premium on Stock X:

$$E(r_X) = \frac{\mathrm{Cov}(r_X, r_M)}{\sigma_M^2} E(r_M)$$

- Risk is measured by covariance with the market, not total variance

CAPM Expected Return–Beta Relationship

- Define beta as:

$$\beta_X = \frac{\mathrm{Cov}(r_X, r_M)}{\sigma_M^2}$$

- Substitute into the equilibrium risk premium expression:

$$E(r_X) = \beta_X \, E(r_M)$$

- Convert back to total expected return:

$$E(R_X) = R_f + \beta_X \big(E(R_M) - R_f\big)$$

- CAPM prediction:
  - systematic risk (beta) is priced
  - idiosyncratic volatility is not

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The Expected Return–Beta Relationship

- From the equilibrium derivation, we obtained:

$$E(R_X) = R_f + \beta_X \big(E(R_M) - R_f\big)$$

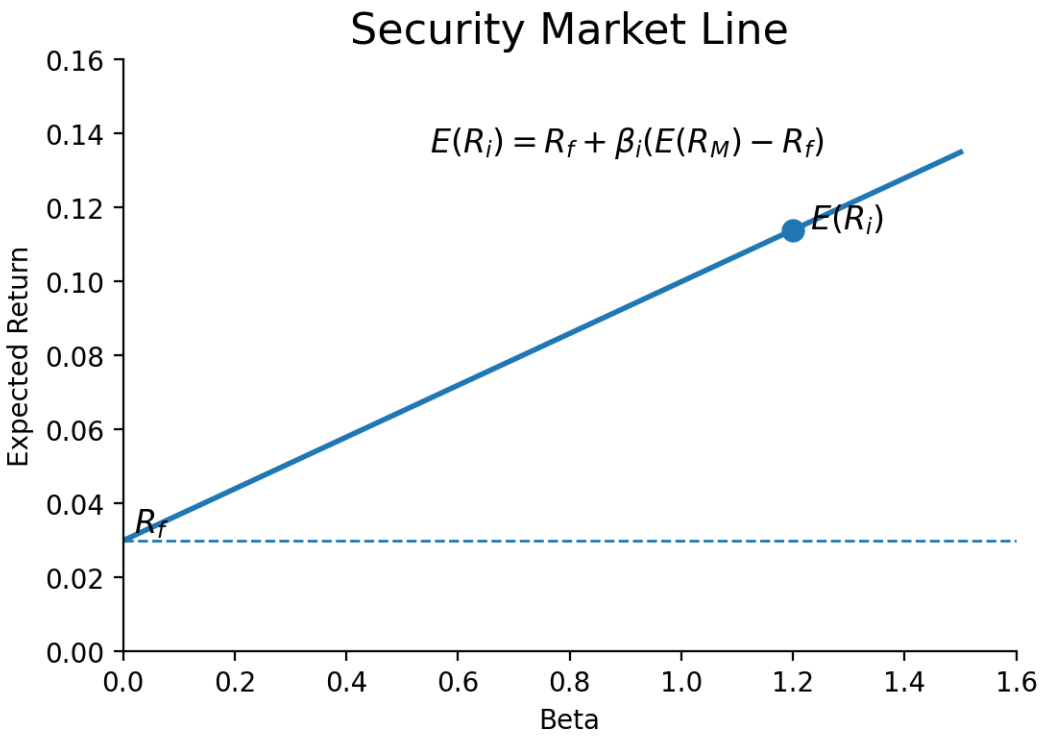
- This equation is the expected return–beta (mean–beta) relationship.
- It decomposes expected return into two components:
  - Time value of money: R_f
  - Risk premium: compensation for bearing systematic risk
- The size of the risk premium depends on:
  - the market risk premium
  - the asset's beta (its contribution to market risk)

Economic Interpretation of Beta

- Beta measures how much an asset contributes to the risk of the market portfolio.
- Assets with high beta:
  - amplify market movements
  - perform poorly in bad states of the world
  - require higher expected returns
- Assets with low or negative beta:
  - provide insurance-like payoffs
  - reduce portfolio risk
  - require lower (or even negative) risk premia
- Total volatility of an asset is irrelevant for pricing

The Security Market Line (SML)

- The Security Market Line is the graphical representation of the expected return–beta relationship.
- Axes:
  - x-axis: beta
  - y-axis: expected return
  - Intercept: R_f
  - Slope: $E(R_M) - R_f$
- Every correctly priced asset must lie on this line in equilibrium.



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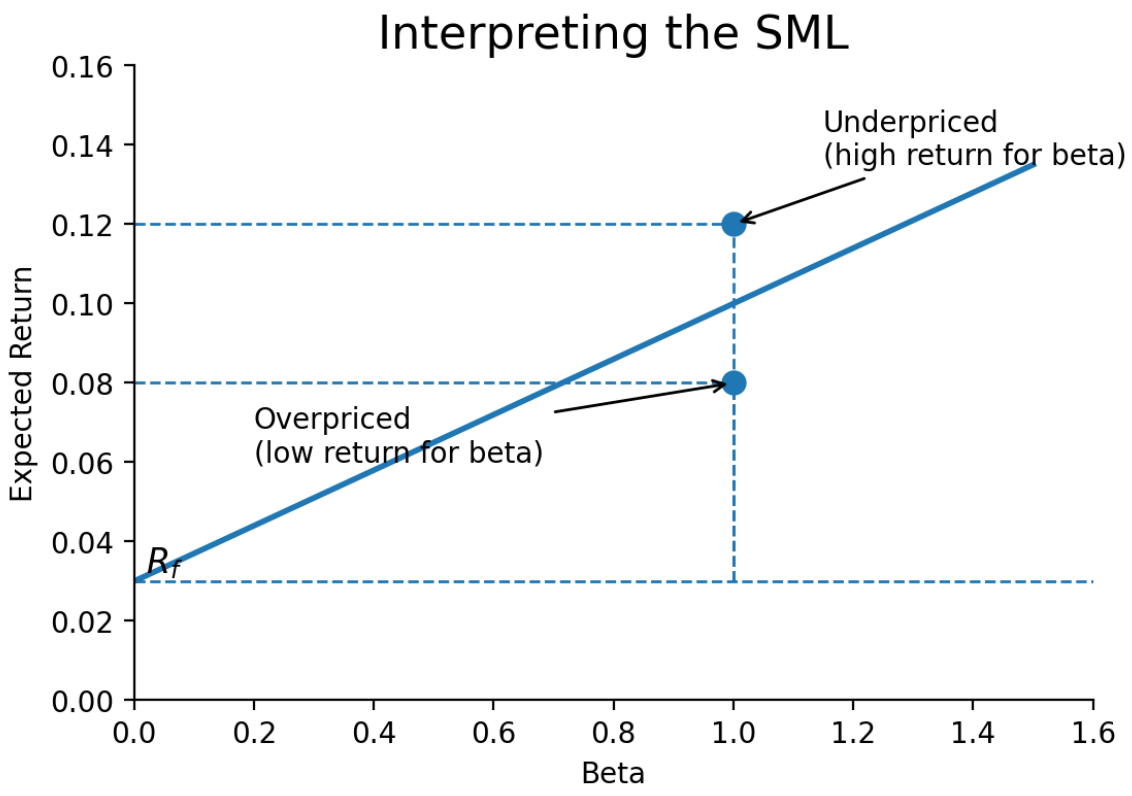
Interpreting the SML

- Assets above the SML:
  - offer too high an expected return for their beta
  - are underpriced
  - will experience buying pressure
- Assets below the SML:
  - offer too low an expected return for their beta
  - are overpriced
  - will experience selling pressure
- Price adjustments continue until assets move back onto the SML.



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The SML as a Benchmark

- The SML provides a benchmark required return for any risky asset.
- Applications:
  - evaluating individual securities
  - estimating a project's cost of capital
  - assessing portfolio manager performance (alpha)
- Alpha is defined as:

$$\alpha_X = E(R_X) - [R_f + \beta_X(E(R_M) - R_f)]$$

- In equilibrium:
  - expected alpha is zero
  - persistent positive alpha is difficult to achieve

Empirical Implementation: Estimating Beta

- In practice, beta is estimated using a market index proxy and historical returns:

$$r_{X,t} = \alpha_X + \beta_X r_{M,t} + e_{X,t}$$

- Interpretation:
  - β_X :systematic market risk exposure
  - e_X :idiosyncratic shock (diversifiable)
  - Under CAPM equilibrium: expected $\alpha_X = 0$

CAPM Assumptions: Which Matter in Practice?

- Homogeneous Expectations / Inputs
  - CAPM assumes investors broadly agree on:
    - expected returns
    - Variances
    - Covariances
  - At first glance this seems unrealistic:
    - investors process information differently
    - analysts generate different forecasts
    - “input lists” may not coincide exactly
  - However, equilibrium provides a disciplining force:
    - if investors hold systematically different expectations, they will trade
    - trading pressure moves prices until expected returns are consistent with market clearing
    - disagreement can persist, but prices reflect an aggregation of beliefs
  - Since most firm information is public large investors often converge toward similar estimates
  - Thus, CAPM does not require identical forecasts, only that equilibrium prices reflect broadly shared information.

CAPM Assumptions: Which Matter in Practice? (con.)

- Short Selling Constraints
- CAPM allows unlimited short sales.
- In reality, shorting is costly and restricted:
  - collateral and margin requirements
  - limited supply of shares to borrow
  - borrowing fees
  - regulatory restrictions during market stress
- Short-sale constraints weaken arbitrage and can allow mispricing to persist.

CAPM Assumptions: Which Matter in Practice? (con.)

- Taxes and After-Tax Returns: CAPM assumes no taxes.
- Taxes cast doubt on this assumption because:
  - two investors can earn different after-tax returns on the same stock
  - optimal risky portfolios could differ across tax brackets
  - equilibrium expected returns might reflect heterogeneous tax treatment
- A plausible resolution lies in clientele and supply effects:
  - high-tax investors may avoid high-dividend stocks, pushing prices down
  - tax-exempt or low-tax investors then view these stocks as bargains and absorb demand
  - Corporations also respond endogenously:
    - if dividends are heavily taxed, firms may substitute toward share repurchases
    - payout policy adjusts, partially neutralizing tax distortions
- Thus, while taxes complicate CAPM's assumptions, equilibrium clienteles and corporate supply responses may reduce their impact on observed prices.

CAPM Assumptions: Which Matter in Practice? (con.)

- Borrowing and Lending at the Risk-Free Rate
- CAPM assumes unlimited borrowing/lending at R_f
- In reality:
  - borrowing rates exceed lending rates
  - leverage constraints bind for many investors
- These frictions weaken two-fund separation and motivate multiperiod and constrained-market extensions.
- Bottom Line:
  - CAPM remains a useful benchmark
  - But frictions in short selling, borrowing, liquidity, and taxes help explain why real markets deviate from the textbook model.

The Market Portfolio Identification Problem

- CAPM requires the true market portfolio as the benchmark risky asset.
- In theory, the market portfolio includes:
  - publicly traded equities
  - bonds and real assets
  - private businesses
  - human capital (future labor income)
- Many of these assets are not freely traded or observable.
- As a result:
  - investors face different exposures to non-traded risks
  - equilibrium demand for publicly traded assets can differ across investors
- Example:
  - an investor whose labor income is highly correlated with economic growth
  - already holds substantial “implicit” market risk
  - may rationally tilt away from high-beta stocks
- Thus, exposure to non-traded assets can alter equilibrium holdings and complicate empirical implementation of CAPM.

Multiperiod Models and Hedge Portfolios

- CAPM is a single-period model:
  - only end-of-period wealth matters
- In reality, investors care about:
  - future investment opportunities
  - changes in interest rates
  - Inflation
  - labor income risk
- In multiperiod settings, expected returns compensate investors for:
  - market risk
  - changes in state variables affecting future opportunities

Liquidity and the CAPM

- CAPM assumes frictionless trading.
- In practice, liquidity matters:
  - transaction costs
  - market depth
  - price impact
  - funding constraints
- Less liquid assets:
  - are harder to sell in bad times
  - expose investors to liquidity shocks
- Investors demand compensation for liquidity risk:
  - expected returns include a liquidity premium
  - Liquidity-augmented CAPM extends the basic model to incorporate trading frictions.

When CAPM Is Most Useful

- Despite its limitations, CAPM remains valuable as a benchmark when:
  - investors are broadly diversified
  - idiosyncratic risk is negligible
  - the market factor dominates the covariance structure
  - portfolios are benchmarked to broad indices
  - trading costs are small
- In these environments:
  - beta is a reasonable first-order measure of risk
  - the Security Market Line provides a required return benchmark
- CAPM is widely used in practice (asset management and corporate finance)

Empirical Challenges to CAPM

- Empirical evidence shows systematic departures from CAPM:
  - size effect: small stocks earn higher average returns
  - value effect: high book-to-market stocks outperform
  - momentum: winners continue to outperform losers
  - low-beta anomaly: defensive stocks often outperform
- Implications:
  - systematic risk is multidimensional
  - market beta alone does not explain the cross-section of returns
  - CAPM is often rejected empirically

From CAPM to Factor Models

- CAPM implies a single source of priced risk:
  - covariance with the market portfolio
- Empirical evidence suggests multiple systematic risks.
- This motivates a general representation of returns:
  - assets respond to several common economic forces
  - comovement arises from shared factor exposures

The General Factor Model

- A k-factor return model:

$$R_i = E(R_i) + \beta_{i1}F_1 + \beta_{i2}F_2 + \cdots + \beta_{ik}F_k + e_i$$

- where:
  - F_j :common systematic factor shocks
  - β_{ij} :sensitivity of asset i to factor j
  - e_i :idiosyncratic shock (diversifiable)
- Covariance across assets is driven by common factor exposures.

Systematic vs Idiosyncratic Risk in Factor Models

- Total risk decomposes into:
  - factor-driven systematic risk
  - asset-specific idiosyncratic risk
- In large diversified portfolios:
  - idiosyncratic shocks average out
  - factor exposures remain
- Thus, equilibrium expected returns should depend only on factor risk, not firm-specific noise.

What Are the Factors?

- Potential macroeconomic factors include:
  - GDP growth shocks
  - inflation surprises
  - interest rate movements
  - credit spreads
  - liquidity conditions
  - business cycle risk
- Factors must satisfy:
  - they affect many assets simultaneously
  - they cannot be diversified away

Arbitrage Pricing Theory (APT)

- Arbitrage Pricing Theory derives asset prices from the absence of arbitrage, rather than investor optimization.
- Arbitrage is defined as a trading strategy with:
  - zero net investment
  - no exposure to systematic risk
  - a strictly positive expected payoff
- If two assets (or portfolios) have identical factor exposures but different expected returns:
  - investors can form a zero-cost long–short position
  - earn a riskless (or near-riskless) profit
- In equilibrium:
  - arbitrage opportunities cannot persist
  - prices must adjust to eliminate return differences for identical factor risk
- APT therefore implies linear pricing restrictions across assets with shared factor exposures.

Core Assumptions of APT

- APT relies on a small number of economically intuitive assumptions:
  - Asset returns can be represented by a factor structure:
    - common systematic factors
    - asset-specific idiosyncratic shocks
  - Idiosyncratic risk can be diversified away:
    - arbitrage portfolios are well diversified
    - firm-specific shocks average out
  - Markets do not allow persistent arbitrage opportunities:
    - portfolios with the same factor exposures must earn the same expected return
  - No assumptions are required about:
    - investor preferences
    - mean–variance optimization
    - the existence of a market portfolio
  - Thus, APT pricing follows from diversification and no-arbitrage, not from a representative investor.

APT Expected Return Relationship

- Under the absence of arbitrage, expected returns satisfy a linear factor pricing relation:

$$E(R_i) = R_f + \beta_{i1}E(r_{F_1}) + \beta_{i2}E(r_{F_2}) + \cdots + \beta_{ik}E(r_{F_k})$$

- where:
  - β_{ij} : exposure of asset i to factor j
  - $E\left(r_{F_j}\right)$: expected excess return (risk premium) on factor-mimicking portfolio j
- Assets earn higher expected returns if they load on factors with positive expected premia
- Two assets with identical factor exposures must have the same expected return; otherwise, an arbitrage opportunity exists

The APT Pricing Relation ("APT SML")

- CAPM implies one Security Market Line:
  - expected return is linear in market beta
- APT implies a multidimensional pricing relation:
  - expected return is linear in multiple factor betas

$$E(R_i) = R_f + \sum_{j=1}^{k} \beta_{ij} E(r_{F_j})$$

- Each systematic factor contributes:
  - a quantity of risk: β_{ij}
  - a price of risk: $E(r_{Fj})$
- Mispricing arises when:
  - an asset's expected return deviates from that implied by its factor exposures
  - arbitrage portfolios exploit these deviations until prices adjust

CAPM and APT

- APT serves many of the same functions as the CAPM:
  - provides a benchmark for required rates of return
  - useful in valuation, capital budgeting, and performance evaluation
  - highlights the distinction between:
    - systematic (factor) risk, which is priced
    - idiosyncratic risk, which is diversifiable and not priced
- Single-Factor Case: CAPM and APT Coincide
  - If there is only one systematic factor—the market portfolio—then:

$$E(R_i) = R_f + \beta_i E(r_M)$$

  - In this sense, CAPM can be viewed as a special case of a one-factor APT.

CAPM and APT (con.)

- Why APT is appealing
  - APT does not require:
    - all investors to be mean–variance optimizers
    - the true (unobservable) market portfolio of all assets
  - Instead, it is built on:
    - well-diversified portfolios
    - the absence of arbitrage opportunities
    - linear pricing across factor exposures
- But APT does not fully dominate CAPM
  - CAPM provides a sharp equilibrium prediction: the expected return–beta relationship should hold for all securities
  - APT implies this relationship holds broadly:
    - especially for well-diversified portfolios
    - but residual risk may remain for individual assets

What Might the Factors Represent?

- The baseline CAPM focuses on a single source of systematic risk.
- In reality, the economy is exposed to multiple pervasive risks:
  - business-cycle fluctuations
  - inflation surprises
  - interest rate and term-structure movements
  - credit conditions
  - liquidity disruptions
- Securities differ in their exposure to these risks.
- Expected return differences reflect compensation for exposure to these common risks:
  - higher factor exposure means higher required return
  - hedging exposure means lower required return

From APT to Empirical Factor Models

- APT establishes that expected returns are linear in exposures to systematic factors.
- However, APT does not specify:
  - how many factors there are
  - what the factors should be
  - how to identify them empirically
- Thus, APT is a pricing framework, not a complete empirical model.
- To implement APT, we must choose:
  - a set of factor-mimicking portfolios
  - factors that are:
    - Pervasive
    - economically meaningful
    - empirically successful in explaining returns
- Different choices of factors imply different multifactor models.

Two Approaches to Factor Identification

Macroeconomic Factors

- inflation shocks
- industrial production growth
- term spreads and default spreads
- Pros:
  - clear economic interpretation
- Cons:
  - difficult to measure cleanly
  - unstable over time

Return-Based Factor Portfolios

- factors constructed from traded securities
- directly observable and investable
- easy to estimate and test
- This approach has proven most successful empirically.

Motivation: Patterns Beyond the CAPM

- Empirical evidence suggests systematic return differences unexplained by market beta:
  - Size effect: small firms earn higher average returns
  - Value effect: high book-to-market firms outperform
  - Momentum: winners continue to outperform losers
  - Profitability / investment effects: strong firms earn higher returns
- These regularities motivate factor portfolios that capture common return drivers.

The Fama–French Three-Factor Model

- Fama and French propose a practical multifactor model:

$$r_i = \alpha_i + \beta_{iM} r_M + \beta_{iS} r_{SMB} + \beta_{iV} r_{HML} + \varepsilon_i$$

- where:
  - r_M :market excess return
  - r_{SMB} :size factor (Small minus Big)
  - r_{HML} :value factor (High minus Low book-to-market)
- Expected returns are linear in exposures to these traded factor portfolios.

Interpretation of the Factors

- Market Factor (r_M)
  - compensation for broad systematic equity risk
- SMB (Size Factor)
  - captures the historical return premium of small firms
  - often linked to distress, liquidity, or growth-option risk
- HML (Value Factor)
  - captures the return premium of value stocks
  - often associated with financial distress or cyclical sensitivity
- The factors summarize pervasive comovement patterns in returns.

Estimating the Three-Factor Model

- Estimation uses a multivariate linear regression of asset excess returns on factor returns, run in the time series.

$$r_{i,t} = \alpha_i + \beta_{iM} r_{M,t} + \beta_{iS} r_{SMB,t} + \beta_{iV} r_{HML,t} + \varepsilon_{i,t}$$

- Outputs:
  - β's: factor exposures
  - α: abnormal return unexplained by the factors
  - R^2 :explanatory power of the factor model
- If the model prices assets well, estimated alphas should be near zero.

Factor Models as Performance Benchmarks

- CAPM evaluates performance relative to market beta only.
- Multifactor models provide richer benchmarks:

$$\alpha_i = r_i - (\beta_M r_M + \beta_S r_{SMB} + \beta_V r_{HML})$$

- Applications:
  - portfolio attribution
  - risk decomposition
  - manager evaluation
  - cost of capital estimation
- A manager may appear to outperform CAPM simply by tilting toward size or value.

Extensions of the Three-Factor Model

- Evidence suggests additional factors improve explanatory power:
  - Momentum (Carhart 4-factor model)
  - Profitability and investment (Fama–French 5-factor model)
  - Liquidity factors
  - Low-volatility and quality factors
- The factor zoo reflects both:
  - compensation for additional risks
  - possible persistent mispricing

The “Factor Zoo” and Model Selection

- Hundreds of return predictors and proposed factors have been documented:
  - value, size, momentum
  - profitability, investment
  - low volatility, quality
  - carry, liquidity, volatility risk
  - dozens more…
- This proliferation is sometimes called the factor zoo.

Data Mining Concern

- With many candidate factors:
  - some will appear significant by chance
  - in-sample success may not persist out-of-sample
- Modern asset pricing emphasizes:
  - robustness across markets and time
  - economic interpretation
  - implementability after trading costs
- A small set of factors (market, value, size, momentum, profitability) explains much of the cross-section.
- Factor investing should focus on:
  - parsimonious models
  - disciplined portfolio construction
  - avoiding overfitting

Smart Beta and Factor Investing

- Smart beta strategies implement factor tilts systematically:
  - Value
  - Size
  - Momentum
  - Low volatility
  - Quality
- Factor investing bridges:
  - academic asset pricing
  - practical portfolio construction
- The idea is that returns can be decomposed into exposures to rewarded systematic factors rather than “alpha.”

Examples of Smart Beta / Factor-Based ETFs

- **iShares Russell 1000 Value ETF (IWD)** — targets value stocks based on fundamental metrics such as price-to-book and valuation ratios.
- **Vanguard Value ETF (VTV)** — tracks a value-oriented index with criteria based on multiple valuation measures.
- **iShares S&P 500 Growth ETF (IVW)** and iShares Russell 1000 Growth ETF — tilt toward stocks with growth characteristics (e.g., earnings growth).
- **Vanguard Dividend Appreciation ETF (VIG)** — focuses on companies with a record of increasing dividends (another form of factor tilt).
- **Minimum Volatility & Low-Vol ETFs** — such as ETFs that overweight stocks with lower historical volatility, targeting defensive factor exposure.
- **Multi-Factor Smart Beta ETFs** — combine exposures to value, quality, momentum, size, etc., in a single fund (e.g., multi-factor products offered by major issuers).

Interpreting Factor Premia: Risk vs. Mispricing

- Multifactor models (e.g., Fama–French) describe systematic return patterns:
  - value stocks outperform growth
  - small stocks outperform large
  - momentum persists
- A central unresolved question is:
  - Do factor premia represent compensation for risk, or persistent mispricing?

Risk-Based Interpretation

- Factor portfolios proxy for exposure to adverse economic states:
  - distress risk
  - cyclical sensitivity
  - liquidity shocks
- Investors require higher expected returns for holding assets that perform poorly in bad times:

$$E(r_{F_j}) > 0 \quad \text{if factor } j \text{ is undesirable risk}$$

- Under this view factor premia are equilibrium risk compensation.

Mispricing / Behavioral Interpretation

- Factor premia may reflect systematic investor biases:
  - overreaction and underreaction
  - extrapolation of growth
  - limits to arbitrage
- Mispricing can persist because:
  - short-sale constraints
  - transaction costs
  - funding risk
- Under this view factor returns represent partially correctable inefficiencies.